Eliminative Information in a Generalized Monty Hall Problem IV

Multiple Prizes with Unequal Rewards

Abstract

Parts I–III developed the generalized Monty Hall problem from identical prizes, to variable reveal costs, to heterogeneous rewards and labeled priors. This note isolates the intermediate case of multiple prizes with unequal values. There are K doors, exactly $m \geq 1$ prize doors, and prize values $v_1, \ldots, v_m$, which need not be equal. We first study the exchangeable model in which the multiset $\{v_1, \ldots, v_m, 0, \ldots, 0\}$ is placed uniformly at random among door labels. In that case, unequal rewards still collapse to a single scalar, the total reward mass

$$V = \sum_{i=1}^{m} v_i.$$

Under uniform switching after t zero reveals, expected stay value is V/K and expected switch value is $V(K-1)/[K(K-1-t)]$. Thus unequal prizes alone do not destroy the one-dimensional structure. The real complexity appears only when door labels carry prior information or when switching ceases to be uniform. We formulate the exchangeable theorem, derive the constant-cost threshold, and supplement the theorem with empirical evidence in two stages: Stage 1 verifies the exchangeable collapse numerically, while Stage 2 shows how label-specific priors break that scalar picture and make initial choice, switching rule, and Monty's policy strategically relevant.

1 Setup

There are K doors, with $1 \leq m < K$ prize doors. Prize door i carries value $v_i > 0$, and the remaining $K - m$ doors carry value 0. Let

$$V = \sum_{i=1}^{m} v_i.$$

We assume throughout that Monty knows realized prize locations and reveals only zero-value doors, never opening the player's initial door.

The player:

- chooses one door initially;
- observes t zero-value reveals, with

$$0 \leq t \leq K - m - 1;$$

- then either stays, or switches to one of the remaining unopened non-initial doors.

The main question is: how much of the Part I and Part III structure survives when there are *multiple prizes with unequal values*?

2 Two Distinct Models

As in Part III, the first split is conceptual.

Exchangeable unequal prizes. The multiset

$$\{v_1, \ldots, v_m, 0, \ldots, 0\}$$

is placed uniformly at random among the K door labels. Door labels carry no prior information.

Labeled unequal prizes. Door labels have distinct priors over reward, so the identity of the initial choice and the identity of Monty's reveals matter.

This note focuses first on the exchangeable case, because it is the cleanest and pins down what unequal prizes do *not* change.

3 Exchangeable Unequal Prizes

Assume the values $v_1, \ldots, v_m$ are fixed, but their placement among the K doors is uniform over labels. The player chooses an initial door uniformly. Monty reveals t zero-value doors. The player then switches uniformly among the remaining unopened non-initial doors.

Theorem 1 (Collapse to total reward mass)**.** *Under exchangeable placement and uniform switching,*

$$\mathbb{E}[\textit{stay reward}] = \frac{V}{K}$$

and

$$\mathbb{E}[\textit{switch reward after } t \textit{ reveals}] = \frac{V(K-1)}{K(K-1-t)}.$$

Proof. By exchangeability, every door has the same ex ante expected reward. Since the sum of rewards over all doors is V, the expected reward of the initial door is V/K.

The expected reward mass outside the initial door is therefore

$$V - \frac{V}{K} = \frac{V(K-1)}{K}.$$

Monty reveals only zero-value doors, so this expected outside reward mass is not altered by the reveals. After t reveals, the remaining non-initial closed set has size $K-1-t$, and symmetry implies that uniform switching spreads this outside reward mass evenly across those switch targets. Hence

$$\mathbb{E}[\text{switch reward after } t \text{ reveals}] = \frac{1}{K-1-t} \cdot \frac{V(K-1)}{K} = \frac{V(K-1)}{K(K-1-t)}.$$

□

Remark 1. *If all positive prizes are unit prizes, then $V = m$, and this reduces exactly to Part I:*

$$\mathbb{E}[\textit{stay}] = \frac{m}{K}, \qquad \mathbb{E}[\textit{switch after } t] = \frac{m(K-1)}{K(K-1-t)}.$$

So unequal prize values do not change the exchangeable theory except through the single scalar V.

4 Advantage of Switching

The switch advantage in the exchangeable unequal-prize model is

$$\frac{V(K-1)}{K(K-1-t)} - \frac{V}{K} = \frac{Vt}{K(K-1-t)}.$$

Thus the sign and comparative statics are exactly the same as in Part I:

- the switch advantage is nonnegative and strictly positive for $t > 0$;
- the gain from switching increases with reveal depth;
- unequal prize values merely rescale the payoff level by V.

This makes the exchangeable unequal-prize model mathematically neat, but not yet qualitatively new.

5 Constant Reveal Costs

Suppose each reveal costs a constant amount c. Then the net switch value after t reveals is

$$W(t) = \frac{V(K-1)}{K(K-1-t)} - ct.$$

Its one-step increment is

$$\Delta_t = W(t+1) - W(t) = \frac{V(K-1)}{K(K-2-t)(K-1-t)} - c.$$

Because the benefit term is increasing in t, the same endpoint logic as in Part I survives. In particular, the comparison between full reveal and no reveal is

$$\frac{V(K-1)}{Km} - c(K-m-1) \quad \text{versus} \quad \frac{V}{K}.$$

This yields the critical constant-cost threshold

$$c^* = \frac{V}{Km}.$$

Remark 2. *When all prizes are unit prizes, $V = m$, so the threshold becomes*

$$c^* = \frac{1}{K},$$

recovering Part I exactly.

6 Stage 1 Empirical Evidence

Stage 1 coding verifies that the exchangeable unequal-prize theorem is not just a symbolic rescaling. The simulation draws explicit prize vectors, places them uniformly among doors, reveals only legal zero doors, and estimates both stay and switch values from Monte Carlo runs.

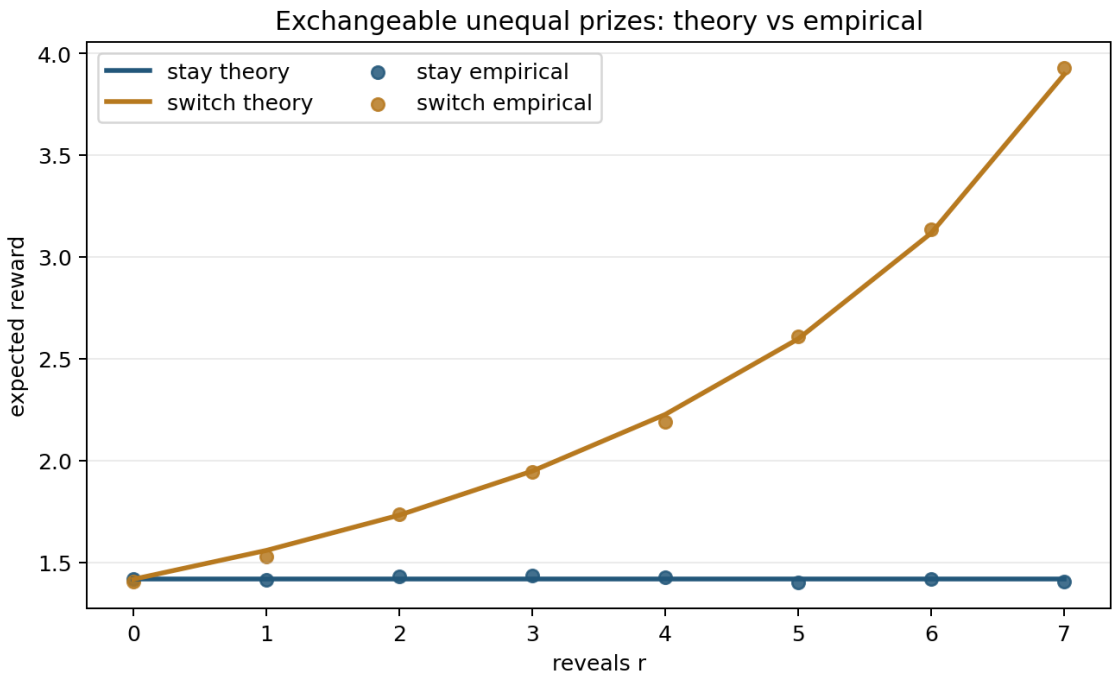

Figure 1: Stage 1 baseline verification for $K = 12$, $m = 4$, and reward vector $(1, 2, 5, 9)$. Stay value is flat at V/K, while switch value rises with reveal depth exactly as predicted by $V(K-1)/[K(K-1-t)]$. The empirical points track the theoretical curves closely.

For the baseline experiment, we fix $K = 12$, $m = 4$, and use the unequal prize vector $(1, 2, 5, 9)$, so $V = 17$. Figure 1 compares the empirical estimates against the exact formulas over all feasible reveal depths $t = 0, 1, \ldots, K - m - 1$. The agreement is close throughout. In the current Stage 1 run, the largest absolute stay error is about 0.017 and the largest absolute switch error is about 0.037, both well within ordinary Monte Carlo noise for a finite simulation of this size.

The more important question is whether the collapse to total reward mass V can be seen directly in simulation when the prize vector itself changes shape. Stage 1 therefore compares three distinct reward vectors,

$$(1, 2, 5, 9), \qquad (4, 4, 4, 5), \qquad (8, 4, 3, 2),$$

all of which satisfy $V = 17$. These vectors differ substantially in how reward mass is distributed across the $m = 4$ prize doors, but under the exchangeable model they should produce the same expected stay and switch values.

Figure 2 shows exactly that. When plotted in raw reward units, the empirical curves for same-V reward vectors lie nearly on top of one another, and they all follow the same theoretical shape. Figure 3 then divides reward by total mass V, making the scale-free claim visually explicit: after normalization, the curves collapse onto the same geometry.

These plots matter because they rule out a tempting but false intuition. One might suspect that a "spikier" prize vector such as $(1, 2, 5, 9)$ should behave differently from a flatter vector such as $(4, 4, 4, 5)$ even in the exchangeable case. Stage 1 shows that this is not so. As long as labels are uninformative and switching is uniform, the simulation respects the theorem: only total reward mass matters.

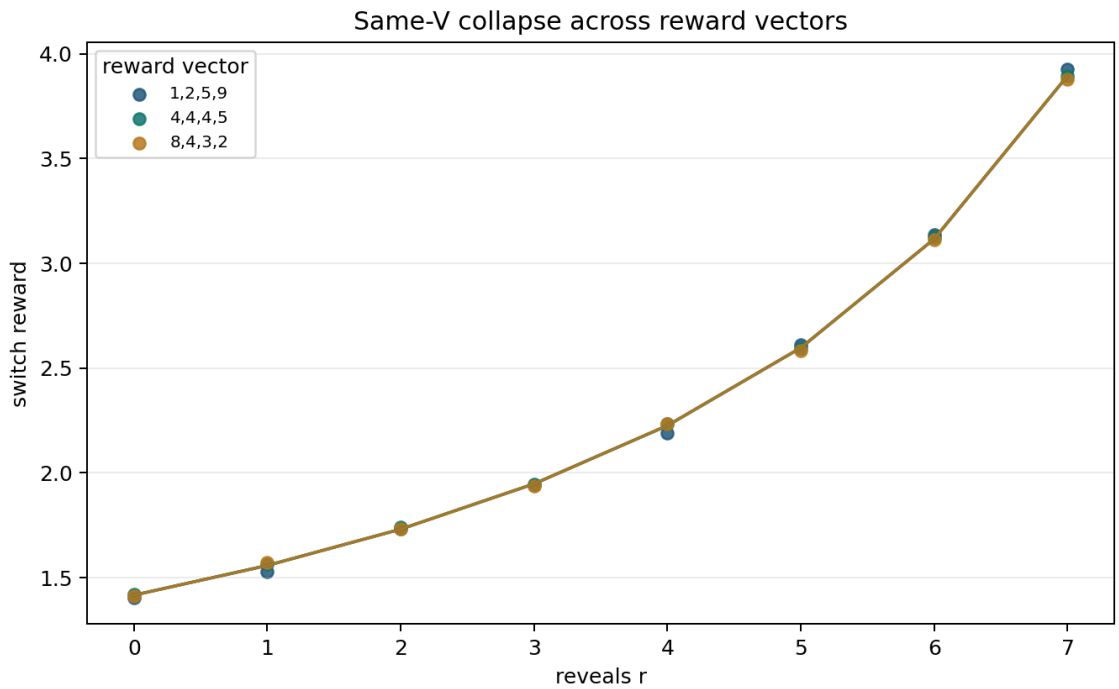

Figure 2: Same-V collapse in raw reward units. Three distinct unequal prize vectors with the same total mass $V = 17$ produce nearly identical empirical stay and switch curves across reveal depth.

7 What Unequal Prizes Do Not Yet Change

The exchangeable unequal-prize model leaves the problem essentially one-dimensional. The relevant state variable is still just the reveal count t, because label symmetry survives. The consequences are:

- stay and switch values depend only on total reward mass V;
- the exact success-law geometry from Part I survives after a scalar rescaling;
- the variable-cost analysis from Part II also carries over with m replaced by V in the relevant payoff scale;
- there is still no strategic initial choice, because all door labels are ex ante equivalent.

So unequal prizes by themselves do *not* create a new decision theory. They create a new payoff scale.

8 Where the Genuinely New Problem Begins

The interesting Part IV problem is therefore not merely "multiple prizes with unequal rewards." It is:

> multiple unequal rewards *with labeled prior information over doors.*

Once labels matter, the state ceases to be scalar. The player must track a posterior reward landscape over remaining doors. Then:

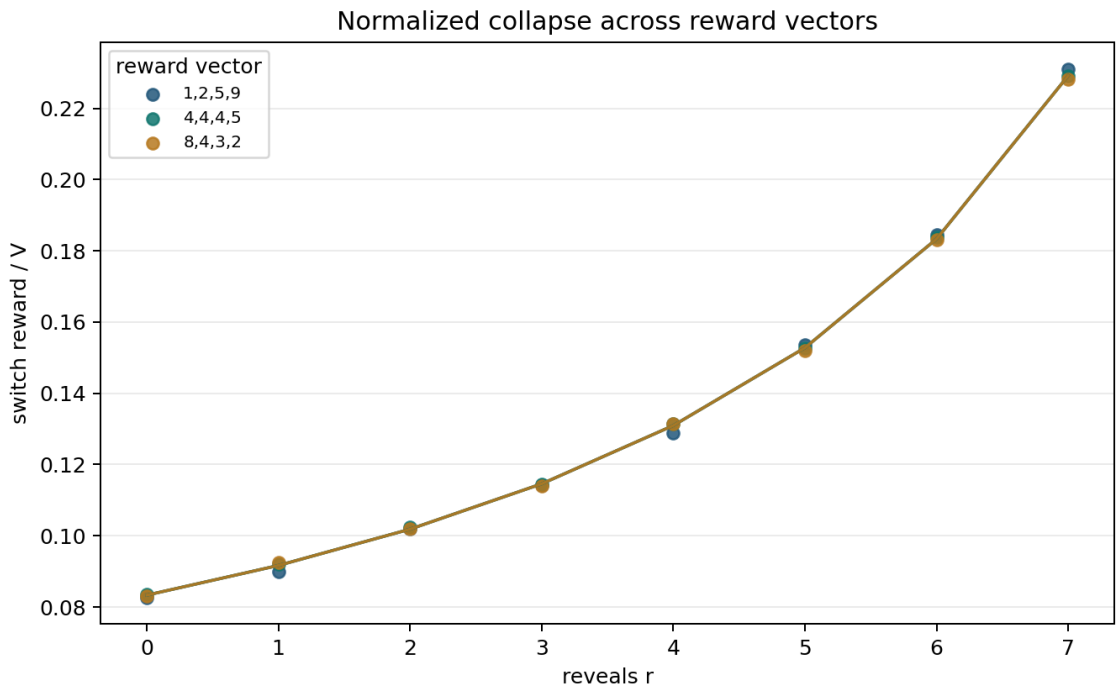

Figure 3: Normalized collapse after dividing by total reward mass V. This is the clearest empirical expression of the theorem: once values are normalized by V, the exchangeable unequal-prize model has the same reveal-depth geometry regardless of how the positive reward mass is partitioned among the m prize doors.

- the initial door can again become strategic;
- different switch rules no longer coincide;
- Monty's reveal policy can carry value-relevant information;
- multiple prize values create a richer posterior allocation problem than in Part III.

A natural first labeled model is:

$$X_j \in \{0, v_j\}$$

with

$$\mathbb{P}(X_j = v_j) = p_j, \qquad \mathbb{P}(X_j = 0) = 1 - p_j,$$

so each door has prior mean

$$\mu_j = p_j v_j.$$

This is the direct multiple-prize unequal-value analogue of the Bernoulli-value model from Part III.

9 Stage 2: Labeled Unequal Prizes

Stage 2 moves from exchangeable reward placement to explicit labeled priors. The point is no longer to verify a scalar collapse theorem, because there is no such scalar collapse left. Instead the goal is to expose which strategic dimensions become relevant once door labels carry information.

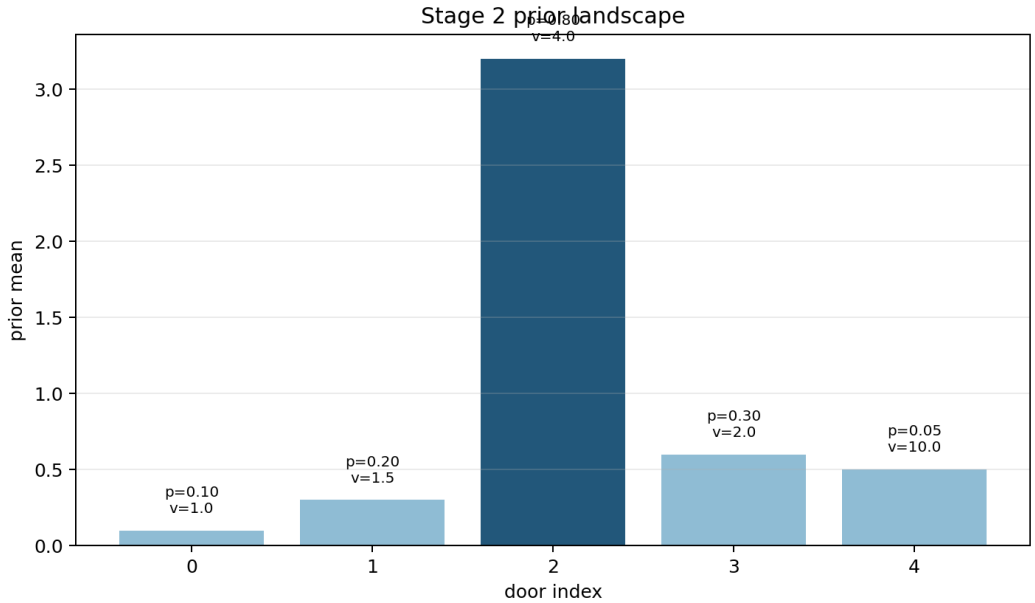

Figure 4: Stage 2 labeled-prior landscape used in the canonical skewed example. The bars show prior means $\mu_j = p_j v_j$, while the annotations show the underlying prize probabilities and reward magnitudes.

The labeled model used in the code is:

$$X_j = \begin{cases} v_j & \text{with probability } p_j, \\ 0 & \text{with probability } 1 - p_j. \end{cases}$$

So each door has prior mean

$$\mu_j = p_j v_j.$$

This model supports three initial strategies ('random', 'highest_mu', 'lowest_mu'), three Monty policies ('uniform_zero', 'low_mu_zero', 'high_mu_zero'), and four switch rules ('stay', 'uniform_switch', 'prior_best_switch', 'oracle_best_switch').

9.1 A canonical skewed prior landscape

For the first Stage 2 experiment, we use the explicit door priors

$$(p_j, v_j) \in \{(0.1, 1),\ (0.2, 1.5),\ (0.8, 4),\ (0.3, 2),\ (0.05, 10)\}.$$

The resulting prior means are

$$\mu = (0.1,\ 0.3,\ 3.2,\ 0.6,\ 0.5).$$

Figure 4 shows this landscape directly. There is one very strong prior-best door, several middling alternatives, and one rare-but-large reward door. This is the sort of structure where label information can plausibly matter.

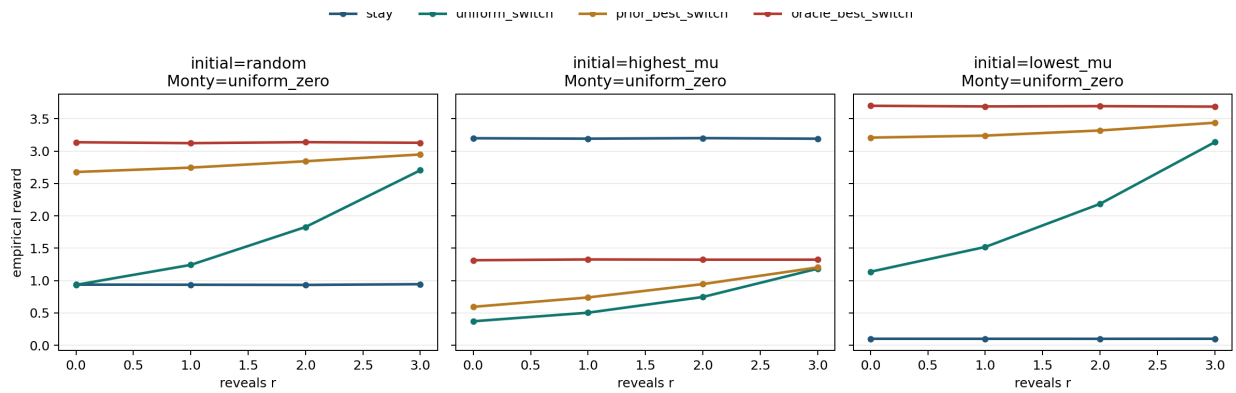

Figure 5: Stage 2 strategy panel under Monty='uniform_zero'. Once labels matter, the initial door and switching rule interact strongly. In this skewed example, 'prior_best_switch' improves on 'uniform_switch' for random and low-value starts, while 'stay' dominates when the player begins at the strongest prior-mean door.

9.2 What Stage 2 changes qualitatively

The first qualitative change is that initial choice now matters sharply. Figure 5 plots empirical reward against reveal depth for all four switch rules under the neutral Monty policy 'uniform_zero', with one panel for each initial strategy.

Three facts stand out.

- If the player starts at the highest-mean door, then staying is hard to beat, because the initial pick already captures most of the prior value.
- If the player starts randomly or at the lowest-mean door, switching becomes attractive, and 'prior_best_switch' can beat 'uniform_switch'.
- 'oracle_best_switch' remains a useful upper bound: even with labeled priors, payoff-relevant uncertainty survives after the reveals.

The second qualitative change is that the comparison between switching rules is now context-dependent rather than universal. In the exchangeable model, uniform switching was analytically forced by symmetry. In the labeled model, it becomes just one rule among several. Sometimes it is too naive, because it ignores useful prior structure. Sometimes it is surprisingly competitive, because the best-looking prior-mean door is not always the door most likely to survive the reveal process as the best remaining option.

This is precisely why the labeled model is genuinely richer than a simple replacement of m by V. The relevant object is no longer a scalar summary, but a posterior landscape over surviving doors.

9.3 Monty policy sensitivity

Stage 2 also lets us ask whether Monty's legal reveal policy materially changes the player's value. Figure 6 summarizes the gain

$$\texttt{prior_best_switch} - \texttt{uniform_switch}$$

at the largest reveal depth used in the experiment, across all initial strategies and all three Monty policies.

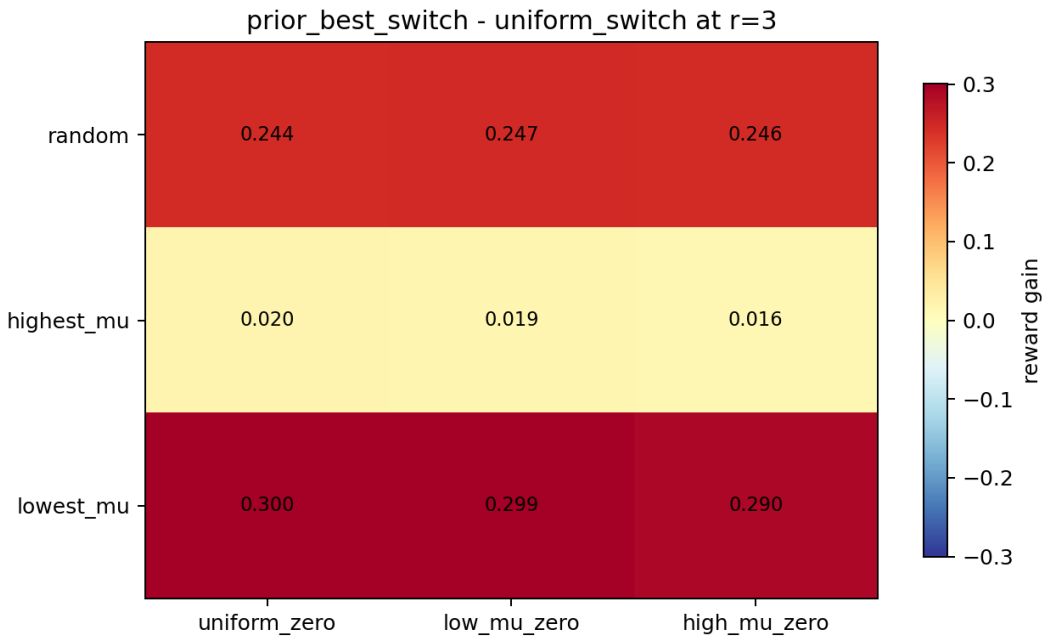

Figure 6: Gain of 'prior_best_switch' over 'uniform_switch' at the largest reveal depth in the Stage 2 run. Positive values show where label-aware switching beats uniform switching. The pattern depends strongly on the initial strategy and only weakly on Monty's reveal bias in this particular skewed example.

In this example, initial choice matters more than Monty's reveal bias: the heat map is structured mainly by the row corresponding to the initial strategy, not by the column corresponding to Monty's legal policy. That need not be true in general, but it is already enough to establish the main Stage 2 point:

> once doors are labeled, the decision problem is genuinely strategic, not just a scalar reveal-depth optimization.

10 Why the Exchangeable Collapse Fails in Stage 2

The Stage 1 collapse worked because the exchangeable model had two symmetry properties:

- all door labels were ex ante identical;
- uniform switching was equivalent to switching by any other label-blind rule.

Those symmetries disappear in Stage 2. Once door j has its own prior mean $\mu_j = p_j v_j$, the player is no longer averaging over an unlabeled reward multiset. The value of a reveal now depends on *which* door was initially chosen, *which* zero doors remain revealable, and *which* candidate switch doors survive.

That is why no single scalar such as total reward mass V can organize the labeled model. In Stage 2, the relevant object is the posterior geometry of the surviving doors. Two landscapes with the same total prior mass can behave quite differently if one has a single dominant door while the other spreads mass more evenly across several alternatives.

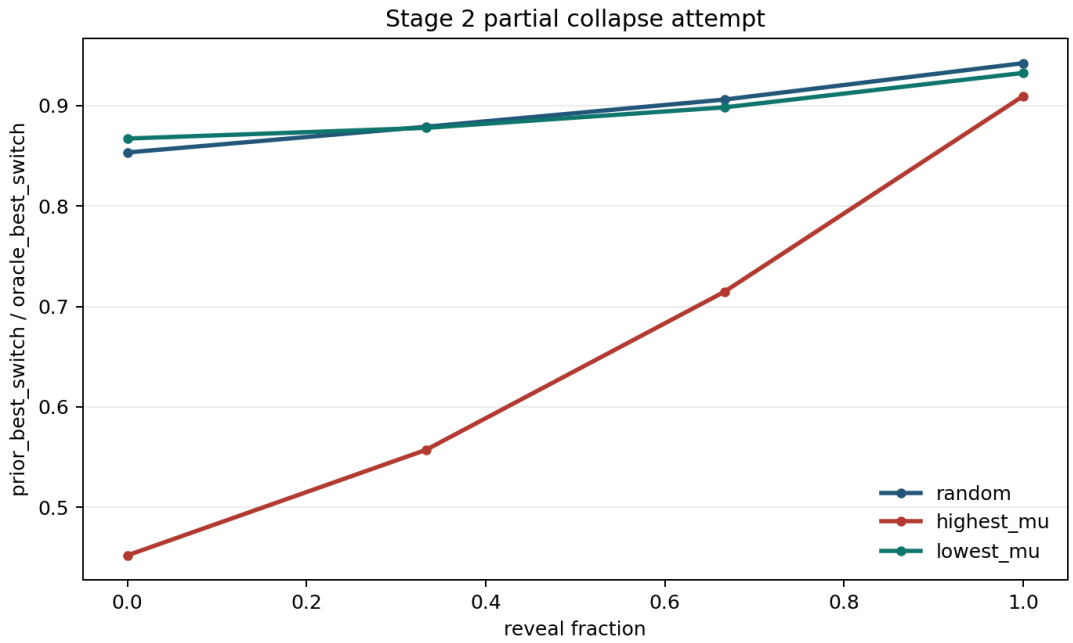

Figure 7: A weaker Stage 2 collapse attempt. The plotted quantity is `prior_best_switch/oracle_best_switch` against reveal fraction, under Monty='uniform_zero'. The curves do not collapse fully, but the 'random' and 'lowest_mu' starts track each other much more closely than either tracks the 'highest_mu' start.

11 A Weaker Collapse Candidate

Although the strong Stage 1 collapse fails, one can still ask whether a weaker rescaling survives. A natural exploratory choice is to compare a feasible label-aware policy against the Stage 2 upper bound:

$$\frac{\texttt{prior_best_switch}}{\texttt{oracle_best_switch}}.$$

This removes the raw reward scale and asks instead how efficiently the player's label-aware heuristic tracks the best ex post switch door as reveal depth increases.

Figure 7 plots this quantity against reveal fraction in the canonical skewed example. The result is not a universal collapse. However, it does show a suggestive weaker pattern: the 'random' and 'lowest_mu' starts lie fairly close together, while the 'highest_mu' start remains distinct. So a partial collapse may exist *within classes of initial conditions*, even though the full scalar exchangeable collapse is gone.

This is the right way to think about labeled unequal rewards. The old scalar collapse should not be expected to survive wholesale. The real question is whether there are *conditional* or *class-dependent* collapses after one controls for coarse structural features such as:

- initial-choice class;
- shape of the prior-mean landscape;
- number of near-best alternatives;
- reveal fraction rather than raw reveal count.

12 Next Part IV Questions

With Stage 2 live, the next coding and theory questions are:

1. In the labeled unequal-prize model, when does prior-best switching beat uniform switching by a large margin?
2. Can low-value initial choices preserve a better switch set when there are several high-value prize candidates rather than just one standout door?
3. How sensitive is the switch value to Monty's legal reveal policy?
4. Does a useful reduced state survive, or does the problem become genuinely high-dimensional in the posterior reward vector?

These are the questions that would make Part IV mathematically distinct from Part III.

13 Conclusion

Multiple prizes with unequal rewards split cleanly into two regimes.

In the exchangeable regime, unequal rewards collapse to total reward mass V. The theory remains scalar, and the Part I and Part II arguments survive with only a rescaling of value. The new Stage 1 evidence shows that this is not just a formal theorem: simulation produces the same reveal-depth curves for very different unequal prize vectors once total reward mass is held fixed.

In the labeled regime, that scalar picture breaks. Stage 2 now makes this live rather than conjectural: initial choice matters, switching rules separate, and Monty's legal reveal policy can change the payoff surface. The problem is no longer organized by a single quantity like t or V, but by how posterior reward mass is distributed across surviving doors.

The right continuation of Part IV is therefore not to search for a naive extension of the Stage 1 collapse, but to identify what weaker coarse-grained descriptions might survive in the labeled model. The partial-collapse plot in Figure 7 suggests one possible direction: collapses may reappear only after conditioning on structural classes such as the initial-choice regime or the shape of the prior landscape.